Control Unit Architecture & Hardwired/Microprogrammed Control

Q. What is the role of the Control Unit? Differentiate between Hardwired Control Unit and Microprogrammed Control Unit.

> **Definition to Remember**

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1. Hardwired Control Unit

The control logic is implemented physically using digital hardware circuits (logic gates, flip-flops, decoders).

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2. Microprogrammed (Soft-Wired) Control Unit

The control signals are generated by executing a tiny, low-level program (a **Microprogram**) stored in a special, fast memory called **Control Memory (ROM)**.

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3. Key Differences

| Feature | Hardwired CU | Microprogrammed CU |

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> **Must-Write Points (for 10 marks)**

> 1. **

> **Quick Recall**

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